Project Kavach: Pure-Software Indic Cyber-Fraud Defense Platform

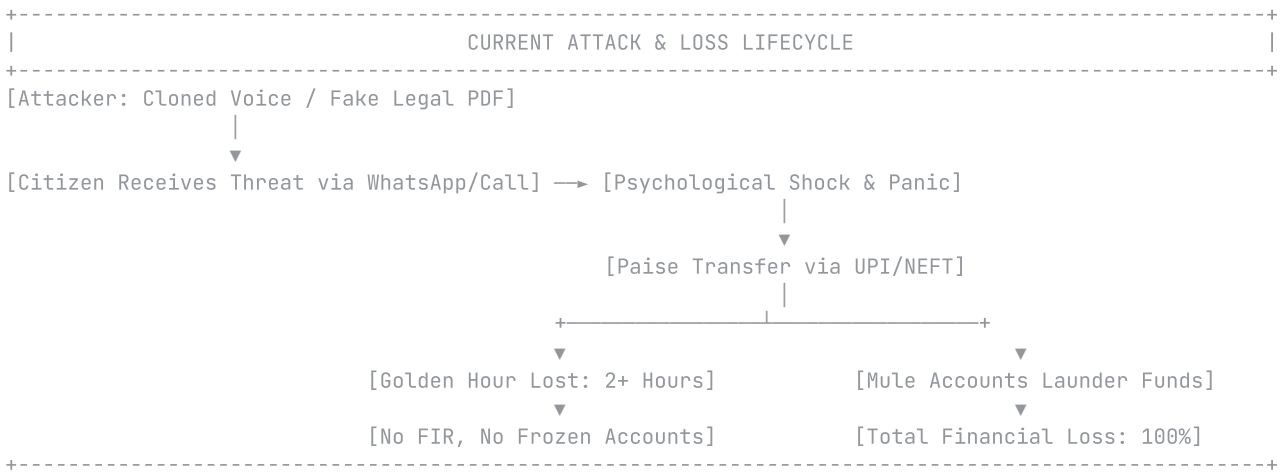
Complete Technical Architecture, System Requirements & Commercialization Roadmap (Hinglish Edition)

1. Executive Summary & Problem Breakdown (Asli Problem Kya Hai?)

India me cyber fraud ab basic lottery SMS ya fake OTP calls tak seemit nahi raha. Scammers high-scale, psychologically weaponized aur automated techniques use kar rahe hain jo seedha aam nagrikon aur business operations ko destabilize karti hain.

1.1 Major Threat Vectors (Scammers Ka Tareeka)

- **"Digital Arrest" Extortion Syndicates:** Organised cartels victims ko video call (Skype/WhatsApp) par connect karke fake CBI, ED, State Police ya Supreme Court ke rooms create karte hain. Wo victim ko fake arrest warrants, official letterheads aur fabricated court stamps dikhate hain aur ghanton tak call par bandhe rakhte hain jab tak victim apni sari life savings "verification account" me transfer na kar de.
- **Neural Voice Cloning & Impersonation:** Scammers social media reels ya shorts se 5–10 seconds ka audio sample utha kar Generative AI (Zero-Shot TTS) se voice clone kar lete hain. Fir parents ya doston ko call karke bolte hain: *"Bhaiya ka accident ho gaya hai, turant 50,000 bhejo."* Normal call filters ise normal call samajhkar pass kar dete hain.
- **Indic Language & Dialect Masking:** Scammers pure English me baat nahi karte. Wo pure North India me Bhojpuri, Maithili, Haryanvi aur pure Hinglish slang use karte hain. Global cyber protection engines (jaise Norton, McAfee, Cloudflare) Indic multi-lingual semantics samajh nahi pate.
- **"Golden Hour" Panic & Evidence Collapse:** Jab fraud execute hota hai, pehle 60–120 minute sabse crucial hote hain (**Golden Hour**). Agar is time me UTR number, beneficiary UPI ID aur transaction timestamp I4C helpline (1930) par submit kar diya jaye, toh bank funds freeze ho jate hain. Par aam insan panic me aa kar screen capture aur bank SMS collect nahi kar pata, aur paise multi-layer mule accounts me siphon off ho jate hain.



1.2 Solution Mechanism (Hum Kya Bana Rahe Hain?)

Project Kavach ek **100% Cloud-Native Indic Multi-Modal Software Engine** hai jisme do direct entry points hain:

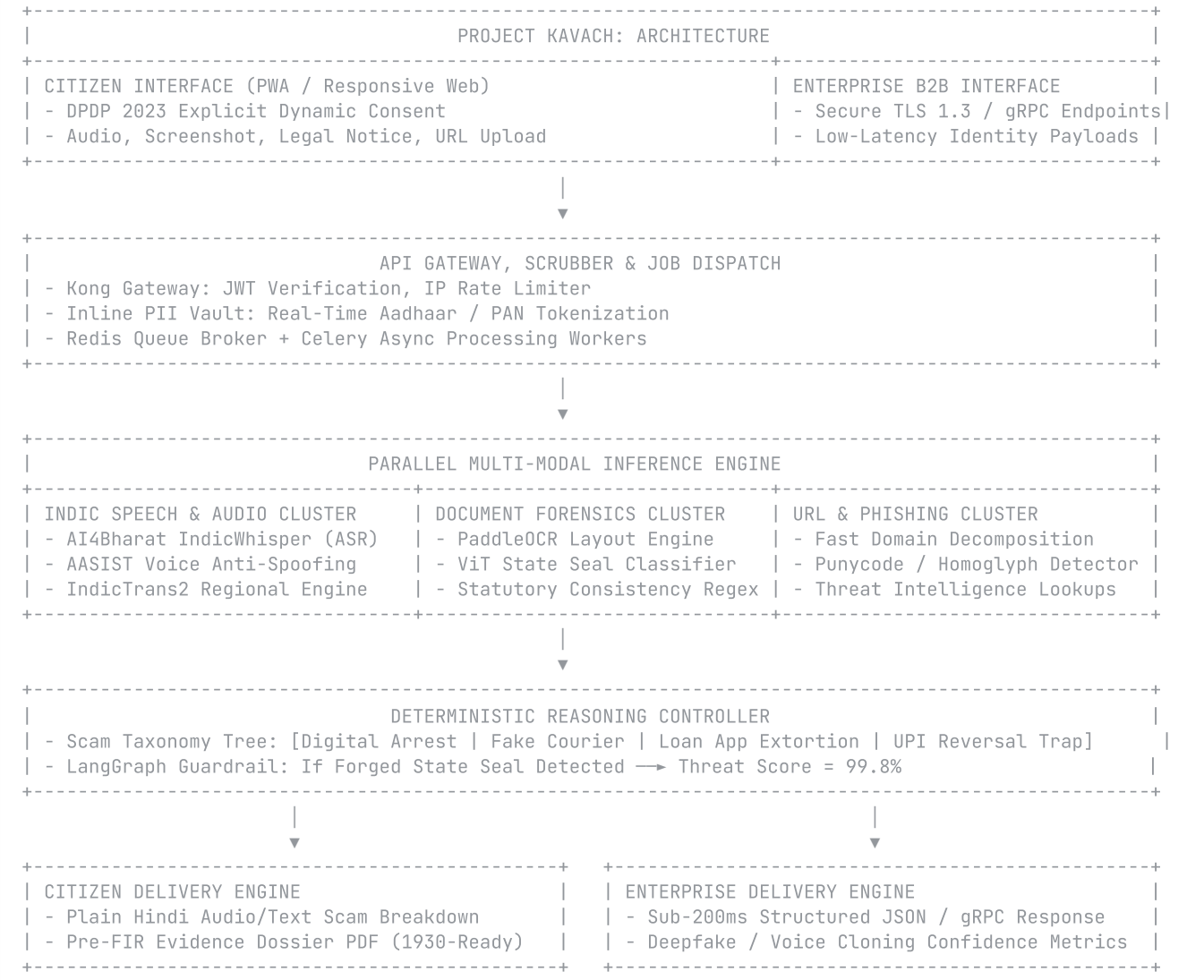
1. **Kavach Citizen Security Web / PWA:** Ek fast, zero-install Progressive Web App jahan citizen koi bhi recording, fake notice ki photo, phishing link ya screenshot forward/upload karega. System turant plain Hindi/Hinglish me scam expose karega aur ek standardized **Pre-FIR Evidence Dossier PDF** bana dega jo direct 1930 ya cybercrime.gov.in me upload ho sake.
2. **Kavach Enterprise B2B API Engine:** Sub-200ms latency wale microservices jo commercial banks, UPI apps (PhonePe, GPay), NBFCs aur telcos ko license kiye jayenge, taaki onboarding (eKYC) aur high-risk fund transfers ke time deepfake voice aur tampered documents real-time block ho sakein.

1.3 Why Hardware Was Scrapped? (Software-Only Decision)

- **TLS 1.3 Encryption Barrier:** Network cables par sara traffic end-to-end encrypted hota hai. Ek hardware firewall (jaise Raspberry Pi) bina har phone me private root certificate install kiye WhatsApp audio ya bank receipts inspect hi nahi kar sakta. Software directly application layer par unencrypted file ko analyze karta hai.
- **Zero Manufacturing Deficit:** Hardware me physical shipment, chip inventory, return logistics aur hardware warranty ka jhanjhat hota hai. Pure software model 85%+ gross margin deliver karta hai.
- **Enterprise Budget Alignment:** Chote kirana dukandar hardware box kharidne ke liye paise nahi dete. Lekin commercial banks fraud rokhne ke liye high-value B2B API contracts sign karte hain.

2. System Architecture & Engineering Blueprint

System ko high-concurrency microservices structure par design kiya gaya hai jo MeitY-empanelled sovereign Indian cloud nodes (AWS Mumbai / Azure Central India) par host hoga.



2.1 Multi-Modal AI Detection Subsystems

- **Indic Speech Pipeline:** AI4Bharat IndicWhisper use karta hai jo raw audio ko text me convert karta hai. Ye model English-Hindi code-mixing (Hinglish) aur regional colloquial terms ko accurately recognize karta hai. Saath hi *AASIST* (Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks) model waveform ke linear prediction cepstral coefficients aur high-frequency phase shifts ko map karke synthetic voice detect karta hai.
- **Document Forensics Engine:** PaddleOCR document ke Devanagari aur English text ko extract karta hai. Ek fine-tuned Vision Transformer (ViT) notice par bane State Emblems, police badges aur stamps ko check karta hai ki wo authentic government designs se match karte hain ya internet se cut-paste kiye gaye hain.

- **Statutory Consistency Regex Engine:** Notice me mention kiye gaye laws ko match karta hai. Scammers aksar purane Indian Penal Code (IPC) ke invalid sections ya fabricated sections likh dete hain. Hamara parser legal discrepancies pakad kar turant alert karta hai.
- **Phishing & Link Deconstructor:** Suspicious links ko parse karke unka age lookup, SSL certificate issuer aur Punycode homoglyphs (jaise SBI ke badle `sb1.com`) analyze karta hai.

3. Functional & Non-Functional Specifications

3.1 Functional Requirements (System Kya Karega?)

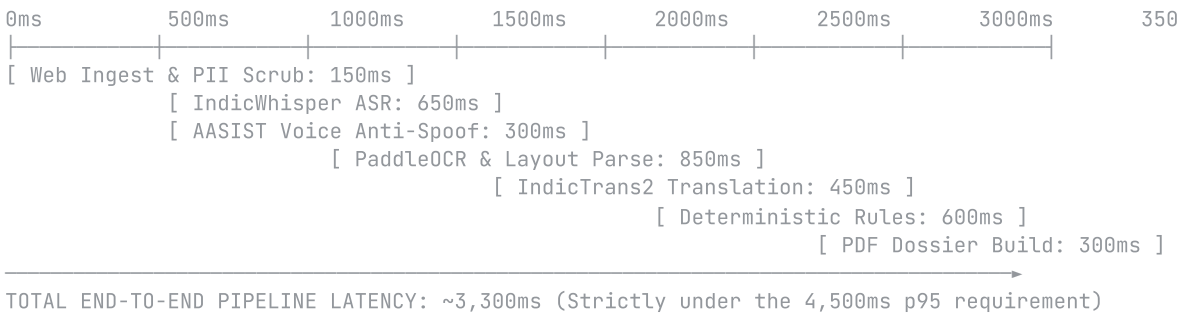
ID	Module Name	Description & Operational Deliverable
FRS-01	Multi-Modal Upload	User voice notes (.wav, .mp3, .ogg), images (.jpeg, .png), legal PDF notices aur URL strings bina app install kiye submit kar sakega.
FRS-02	Synthetic Voice Flagging	Audio stream ko analyze karke synthetic probability score ($P_{\text{synth}} \in [0, 1]$) generate karega aur vocoder artifacts flag karega.
FRS-03	Indic Speech Parser	Hindi, Hinglish aur regional dialects ko readable plain text me transcribe karega with confidence scores.
FRS-04	Fake Seal / Stamp Verifier	Legal notice par Supreme Court, CBI, Police ke stamps ko compare karke forgery verdict dega.
FRS-05	Plain-Hindi Explanation	Jargon-free native Hindi me output generate karega: *"Ye notice nakli hai. Police video call par kisi ko arrest nahi karti."*
FRS-06	Pre-FIR Dossier Builder	Extracted UTR number, fraud UPI ID, beneficiary details aur call timestamps ko standardized PDF format me organize karega.
FRS-07	Enterprise Verification API	Commercial financial apps ke liye sub-200ms document verification aur deepfake detection programmatic endpoints provide karega.

3.2 Non-Functional Requirements (Performance, Security & Legal)

Category	ID	Standard	Validation Criteria
B2B Performance	NFRS-01	Latency ≤ 200 ms	Single-modal synchronous API requests sub-200ms (p95) latency SLA maintain karengi.
Web Performance	NFRS-02	Latency ≤ 4,500 ms	Full multi-modal upload (Audio + OCR + Translation + Synthesis) 4.5 seconds ke andar complete hoga.
System Scale	NFRS-03	5,000 Concurrent Queries	High-load peak bursts ke dauran auto-scale clusters bina packet drop ke load handle karenge.
Data Shredding	NFRS-04	15-Minute Data TTL	Analysis complete hote hi consumer audio aur images cloud RAM aur storage se permanently erase ho jayengi.
Regulatory	NFRS-05	DPDP Act 2023 Strict	Consent records cryptographically audit-logged rahenge aur processing sovereign Indian boundary ke andar hogi.

4. Processing Pipeline & Latency Budget

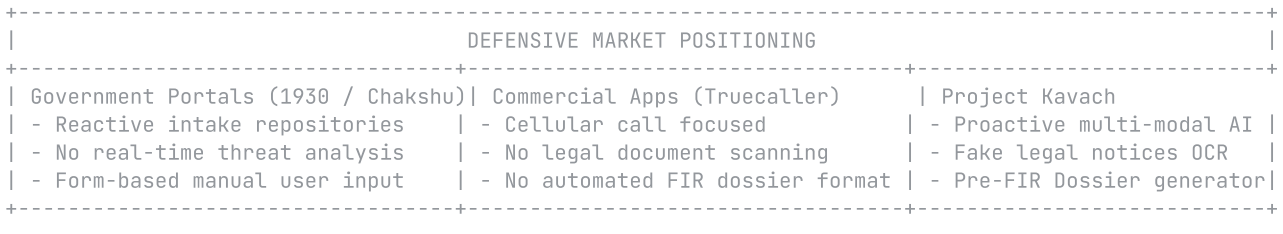
Cloud GPU clusters (NVIDIA L4 Tensor Core instances) par pipeline execution asynchronous micro-queues ke zariye runs hoti hai:



4.1 Pipeline Latency Breakdown Table

Processing Stage	Model / Technology Used	Execution Context	Avg Time (ms)
Payload Ingress & PII Tokenization	Kong Gateway + Crypto Vault	CPU / In-Memory AES-256	150 ms
Speech-to-Text Transcription	AI4Bharat IndicWhisper	NVIDIA L4 GPU (Triton)	650 ms
Synthetic Audio Classification	AASIST Spectro-Temporal GAT	NVIDIA L4 GPU	300 ms
Document OCR & Stamp Classification	PaddleOCR + Vision Transformer	NVIDIA L4 GPU	850 ms
Regional Language Translation	IndicTrans2 (Distilled 200M)	NVIDIA L4 GPU	450 ms
Fraud Taxonomy Matching	LangGraph + Statutory Regex Engine	CPU Microservice	600 ms
Pre-FIR Dossier Synthesis	ReportLab PDF Micro-Engine	CPU Microservice	300 ms

5. Competitive Differentiation vs. Market (Market Me Kya Alag Hai?)



Capability	DoT Sanchar Saathi (Chakshu)	MHA I4C Helpline (1930 / NCRP)	Truecaller AI Call Scanner	Project Kavach
Operational Focus	Citizen fraud reporting to telcos	Bank account lien & capital freezing	Live call spam tagging	Multi-modal forensics & B2B defense
Payload Modality	Text numbers, screenshots	Transaction IDs, account numbers	Cellular voice signals	Audio, legal PDFs, stamps, links
Indic Dialect Depth	Static portal forms	Static call-center triage	Primarily English metadata	Fine-tuned IndicWhisper & IndicTrans2
Document Analysis	None (Manual police review)	None (Manual review post-FIR)	None	Vision Transformer + Judicial seals
Citizen Real-Time Feedback	None (Gives ticket ID)	None (Operator intake)	Binary spam/clone banner	Plain-Hindi root-cause analysis
Complaint Formatting	Manual form filling	Phone call Q&A	None	Instant structured Pre-FIR Dossier
Commercial Model	Public service	Public service	Ads / Consumer subscription	Enterprise B2B API Licensing

6. Financial Architecture & Unit Economics (Paisa Kaise Banega?)

6.1 Why PWA Trumps WhatsApp Business API

WhatsApp Business API har interaction par platform charges lagata hai. India me 1,000 monthly utility messages ke baad har user interaction par Meta charges lagte hain. Agar 10 lakh users Kavach bot par 3–4 messages chat karenge, toh har query par messaging cost plus GPU compute cost mila kar ₹1.50–₹2.50 kharch honge. Mahine ka operational deficit ₹15 lakh se ₹25 lakh ho jayega bina kisi direct revenue ke.

PWA Strategy: Kavach Progressive Web App use karega. Cloudflare CDN ke through web traffic delivery cost sirf ₹0.02 per user aati hai. WhatsApp ko sirf marketing links share karne ke liye use kiya jayega, jisse distribution expense 95% kam ho jata hai.

6.2 Revenue Generation: B2B Enterprise API

- **Commercial Banks & NBFCs:** eKYC verification aur video call onboarding ke waqt deepfake face aur cloned voice verify karne ke liye per-call API rate (₹1.50 se ₹3.00) charge kiya jata hai.
- **UPI & Payment Apps (PhonePe, GPay, Paytm):** Payment transfer screen par recipient handle aur user-uploaded payment screenshot verification API integrate hoti hai.
- **Telecom Security Feeds:** Telecom companies ko real-time Indic scam scripts aur newly emerged malicious domain feeds license kiye jate hain.

6.3 Unit Economics Breakdown

Cost / Revenue Component	Consumer Web Tier (PWA)	Enterprise B2B API Tier
Bandwidth & Network Delivery	₹0.02 (Cloudflare CDN Edge)	₹0.01 (Direct TLS 1.3 / gRPC)
Inference GPU Compute Cost	₹0.85 (Multi-Modal Full Pipeline)	₹0.15 (Single-Check Sub-200ms)
Total Ingress Operational Cost	**₹0.87 per scan**	**₹0.16 per check**
Realized Revenue per Query	₹0.00 (Free Community Tier)	**₹2.00 per API call**
Gross Margin	-₹0.87 (Operational Top-of-Funnel)	**+₹1.84 (92% Gross Margin)**

> **SUSTAINABILITY RATIO**
> Sirf **50,000 paid enterprise API calls** pure platform ke **1,00,000 free citizen scans** ke GPU compute aur infrastructure kharche ko 100% cover kar lete hain.

7. Regulatory, DPDP Act 2023 & Legal Shielding

7.1 DPDP Act 2023 Mandatory Compliance

- **Data Fiduciary Classification:** Jab user apni call recording ya document upload karta hai, Kavach legal terms me Data Fiduciary banta hai. Platform Section 6 ke mutabiq multi-lingual explicit consent screen enforce karta hai.
- **Biometric Data & Purpose Limitation:** Audio me vocal prints biometric data hote hain, aur notice me PAN ya account details hoti hain. Kavach strict *Purpose Limitation* implement karta hai: analysis score

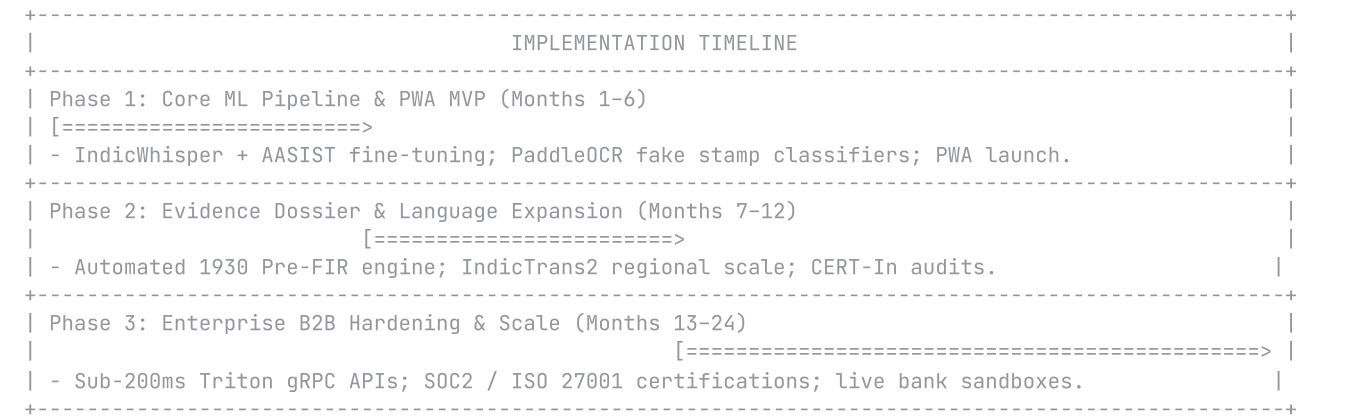
generate hote hi user payload memory se cryptographically shred kar diya jata hai.

- **Penalty Shielding:** DPDP Act ke Schedule 1 ke tahat data breach par ₹250 crore tak ki statutory penalty ho sakti hai. Kavach zero persistent customer storage model par chalta hai aur local Indian data centers me operate karta hai.

7.2 Asymmetric Liability Shield (Court/Police Disclaimers)

- **False Negative Risk:** Agar system kisi naye advanced scam ko clean bol de aur user paise transfer kar de, toh platform par financial loss ka claim na ho sake.
- **False Positive Risk:** Agar system kisi asli court summon ya tax notice ko fake bol de aur citizen court me pesh na ho, toh platform par legal obstruction ka case na bane.
- **Mitigation Strategy:** Output hamesha probabilistic risk assessment ke roop me diya jata hai: *"Risk Rating: 94%. Is document me unauthorized stamps paye gaye hain. Kripya kisi bhi payment se pehle official 1930 helpline par verify karein."* Platform absolute verdict claim nahi karta.

8. Implementation Roadmap (Phases to Build)



8.1 Phase 1: Core ML Pipeline & Web MVP (Months 1–6)

- **Indic Speech Pipeline:** Fine-tune AI4Bharat IndicWhisper on Hindi/Hinglish call datasets; integrate AASIST anti-spoofing audio classifier.
- **Document Forensics:** PaddleOCR pipeline deploy karna with Vision Transformer trained on State police and court emblem datasets.
- **Web PWA Interface:** Deploy lightweight React/Next.js interface with real-time plain-Hindi risk explanations.
- *Deliverable:* Fully functional web app processing audio and documents within 4 seconds.

8.2 Phase 2: Evidence Dossier & Language Expansion (Months 7–12)

- **Pre-FIR Builder:** Automated report generator jo UTRs, UPI IDs aur call records ko structured PDF format me compile karega.
- **Language Scaling:** IndicTrans2 distilled models integrate karke Tamil, Telugu, Bengali aur Marathi support rollout karna.
- **Security Certification:** Comprehensive penetration testing aur CERT-In security empaneled audits complete karna.
- *Deliverable:* Multi-lingual evidence preparation engine active nationwide.

8.3 Phase 3: Enterprise B2B Hardening & Commercialization (Months 13–24)

- **Sub-200ms Optimization:** Serving stack ko NVIDIA Triton Inference Server par re-architect karke gRPC high-throughput enterprise pipelines ready karna.
- **Enterprise Certifications:** SOC2 Type II aur ISO/IEC 27001 audit clearances complete karna.
- **Bank & Fintech Deployments:** Leading commercial banks aur payment gateways ke fraud mitigation sandboxes me API integrate karna.
- *Deliverable:* Enterprise-grade B2B fraud defense infrastructure with long-term recurring SaaS contracts.

9. Final Viability Verdict

- **Original Hardware Model (Kavach Box + Consumer Bot): Unviable.** TLS 1.3 encryption hardware inspection ko block kar deti hai aur WhatsApp messaging bills platform ko loss me daal dete hain.
- **Restructured Pure-Software Model (PWA + B2B API): Highly Viable (9.0/10).** Hardware dependency khatam karke system instant-scale ban jata hai. Zero inventory cost, DPDP Act compliance, aur direct B2B banking monetization is model ko profitable aur defensible banate hain.